

SDS8102: Advanced Mathematical Statistics

Lecture 11: Empirical process

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Empirical Processes, Uniform Convergence, and Complexity

- From pointwise LLN to uniform LLN
- Glivenko–Cantelli classes
- Symmetrization and Rademacher complexity
- VC dimension and generalization
- Covering numbers and chaining
- Donsker classes and uniform CLTs

From LLN to Uniform LLN

Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P.$$

For a measurable function f , write

$$Pf = \mathbb{E}[f(X)], \quad P_n f = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

For a *fixed* f , the LLN gives

$$P_n f \longrightarrow Pf.$$

From LLN to Uniform LLN

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For a *fixed* f , the LLN gives

$$P_n f \longrightarrow Pf.$$

But many statistical problems require

$$\sup_{f \in \mathcal{F}} |P_n f - Pf| \longrightarrow 0.$$

Remark:

The function f is now allowed to vary with the data. Pointwise LLN is therefore not enough.

Why Do We Need Uniform Convergence?

Suppose

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} P_n \ell_f.$$

Even if

$$P_n \ell_f \rightarrow P \ell_f$$

for every fixed f , this says little directly about $\hat{f}$, because $\hat{f}$ is chosen *after observing the data*.

What we really want is

$$\sup_{f \in \mathcal{F}} |(P_n - P) \ell_f| \rightarrow 0.$$

Example:

Uniform convergence appears in:

- empirical CDFs,
- M-estimation,
- empirical risk minimization,
- classification,
- confidence bands,
- nonparametric statistics.

The Empirical Measure

Definition:

The *empirical measure* is

$$P_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}.$$

For every measurable f ,

$$P_n f = \int f dP_n = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

Thus

$$(P_n - P)f = \frac{1}{n} \sum_{i=1}^n \{f(X_i) - Pf\}.$$

The empirical measure P_n is a random approximation to P .

The Empirical Process

Definition:

For a class $\mathcal{F}$, define the empirical process

$$\mathbb{G}_n f = \sqrt{n}(P_n - P)f = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{f(X_i) - Pf\}, \quad f \in \mathcal{F}.$$

There are two natural scales:

$$\sup_{f \in \mathcal{F}} |(P_n - P)f|$$

and

$$\sup_{f \in \mathcal{F}} |\mathbb{G}_n f|.$$

Uniform LLN $\longleftrightarrow$ Glivenko–Cantelli,

Uniform CLT $\longleftrightarrow$ Donsker.

Running Example: The Empirical CDF

For $X_i \in \mathbb{R}$, let

$$F(t) = P(X \leq t), \quad F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{X_i \leq t\}.$$

Define

$$f_t(x) = \mathbb{1}\{x \leq t\}.$$

Then

$$F_n(t) = P_n f_t, \quad F(t) = P f_t.$$

Hence

$$F_n(t) - F(t) = (P_n - P)f_t.$$

So the empirical CDF corresponds to the function class

$$\mathcal{F} = \{\mathbb{1}_{(-\infty, t]} : t \in \mathbb{R}\}.$$

Pointwise versus Uniform Convergence

For every fixed t , the SLLN gives

$$F_n(t) \rightarrow F(t) \quad \text{a.s.}$$

But this only says

$$\forall t, \quad |F_n(t) - F(t)| \rightarrow 0.$$

Glivenko–Cantelli says

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \rightarrow 0 \quad \text{a.s.}$$

Remark:

The order of “sup” and “lim” matters:

$$\forall t, a_n(t) \rightarrow 0 \quad \not\Rightarrow \quad \sup_t |a_n(t)| \rightarrow 0.$$

Theorem (Glivenko–Cantelli)

For the empirical CDF F_n ,

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \longrightarrow 0 \quad a.s.$$

Proof idea:

Fix $\varepsilon > 0$ and choose

$$-\infty = t_0 < t_1 < \cdots < t_m = \infty$$

such that

$$F(t_j) - F(t_{j-1}) \leq \varepsilon.$$

The infinite problem is reduced to controlling finitely many points

$$t_0, \dots, t_m.$$

Proof of Glivenko–Cantelli

If

$$t_{j-1} \leq t \leq t_j,$$

then monotonicity gives

$$F_n(t_{j-1}) \leq F_n(t) \leq F_n(t_j).$$

Therefore

$$F_n(t) - F(t) \leq |F_n(t_j) - F(t_j)| + \varepsilon,$$

and similarly

$$F(t) - F_n(t) \leq |F_n(t_{j-1}) - F(t_{j-1})| + \varepsilon.$$

Hence

$$\sup_t |F_n(t) - F(t)| \leq \max_j |F_n(t_j) - F(t_j)| + \varepsilon.$$

By the SLLN, the finite maximum converges to zero a.s.

Thus

$$\limsup_n \sup_t |F_n(t) - F(t)| \leq \varepsilon.$$

Let $\varepsilon \downarrow 0$.

Glivenko–Cantelli Classes

Definition:

A class $\mathcal{F}$ is called *P-Glivenko–Cantelli* if

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \rightarrow 0 \quad \text{a.s.}$$

The classical GC theorem says

$$\{\mathbb{1}_{(-\infty, t]} : t \in \mathbb{R}\}$$

is GC for every distribution P on $\mathbb{R}$.

The general question is:

What properties of $\mathcal{F}$ guarantee a uniform LLN?

Answer: we need to control the **complexity** of $\mathcal{F}$.

Warm-Up: A Finite Function Class

Suppose

$$\mathcal{F} = \{f_1, \dots, f_N\}, \quad |f| \leq 1.$$

For a fixed f , Hoeffding gives

$$P(|P_n f - Pf| > t) \leq 2e^{-nt^2/2}.$$

By a union bound,

$$P\left(\sup_{f \in \mathcal{F}} |P_n f - Pf| > t\right) \leq 2Ne^{-nt^2/2}.$$

Thus the typical size is

$$\sup_{f \in \mathcal{F}} |P_n f - Pf| \asymp \sqrt{\frac{\log N}{n}}.$$

Remark:

Complexity enters through $\log N$, not N .

What About Infinite Classes?

For an infinite class,

$$|\mathcal{F}| = \infty,$$

so a naive union bound is useless.

We need notions of *effective size*.

Three important measures will appear:

Rademacher complexity	:	ability to fit random noise,
VC dimension	:	combinatorial complexity,
metric entropy	:	size at resolution ε .

These will eventually be connected.

Symmetrization: The Main Difficulty

We want to control

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f|.$$

The difficulty is the unknown population quantity

$$P f.$$

Introduce an independent “ghost sample”

$$X'_1, \dots, X'_n \stackrel{\text{i.i.d.}}{\sim} P,$$

and define

$$P'_n f = \frac{1}{n} \sum_{i=1}^n f(X'_i).$$

Since

$$\mathbb{E}[P'_n f] = P f,$$

we can replace the population mean by a second empirical mean.

Ghost-Sample Symmetrization

Conditional on $X_1, \dots, X_n$,

$$Pf = \mathbb{E}[P'_n f \mid X_1, \dots, X_n].$$

Hence Jensen gives

$$\begin{aligned} \mathbb{E} \sup_f |P_n f - Pf| &= \mathbb{E} \sup_f |P_n f - \mathbb{E}[P'_n f \mid X]| \\ &\leq \mathbb{E} \sup_f |P_n f - P'_n f|. \end{aligned}$$

Therefore

$$\mathbb{E} \sup_f |P_n f - Pf| \leq \mathbb{E} \sup_f \left| \frac{1}{n} \sum_{i=1}^n \{f(X_i) - f(X'_i)\} \right|.$$

The unknown Pf has disappeared.

Rademacher Randomization

Let

$$\varepsilon_1, \dots, \varepsilon_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{-1, +1\}.$$

Since

$$(X_i, X'_i) \stackrel{d}{=} (X'_i, X_i),$$

we have

$$f(X_i) - f(X'_i) \stackrel{d}{=} \varepsilon_i \{f(X_i) - f(X'_i)\}.$$

Thus

$$\begin{aligned} \mathbb{E} \sup_f \left| \frac{1}{n} \sum_i \{f(X_i) - f(X'_i)\} \right| \\ = \mathbb{E} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \{f(X_i) - f(X'_i)\} \right|. \end{aligned}$$

The Symmetrization Inequality

Using the triangle inequality and symmetry of X and X' ,

$$\begin{aligned}\mathbb{E} \sup_f |P_n f - P f| &\leq \mathbb{E} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| \\ &\quad + \mathbb{E} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X'_i) \right|.\end{aligned}$$

Therefore:

Theorem (Symmetrization)

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|.$$

The right-hand side contains only the observed functions and random signs.

Rademacher Complexity

Definition:

The empirical Rademacher complexity is

$$\hat{\mathfrak{R}}_n(\mathcal{F}) = \mathbb{E}_{\varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \middle| X_1, \dots, X_n \right].$$

Its population version is

$$\mathfrak{R}_n(\mathcal{F}) = \mathbb{E}_X \hat{\mathfrak{R}}_n(\mathcal{F}).$$

Symmetrization gives

$$\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2 \mathfrak{R}_n(\mathcal{F}).$$

What Does Rademacher Complexity Measure?

Conditional on the data,

$$\sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_i \varepsilon_i f(X_i) \right|$$

asks:

How well can a function in $\mathcal{F}$ correlate with completely random signs?

- If $\mathcal{F}$ is very rich, it can adapt to the signs.
- If $\mathcal{F}$ is simple, random signs are difficult to fit.

ability to fit noise $\approx$ statistical complexity.
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This is why Rademacher complexity naturally controls generalization.

Application: Empirical Risk Minimization

Let

$$R(f) = \mathbb{E}[\ell(f(X), Y)]$$

and

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i).$$

Let

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{R}_n(f), \quad f^* \in \arg \min_{f \in \mathcal{F}} R(f).$$

Then

$$R(\hat{f}) - R(f^*)$$

is the *excess risk*.

How can uniform convergence control it?

Uniform Convergence $\Rightarrow$ Generalization

Decompose:

$$\begin{aligned} R(\hat{f}) - R(f^*) &= R(\hat{f}) - \hat{R}_n(\hat{f}) \\ &\quad + \hat{R}_n(\hat{f}) - \hat{R}_n(f^*) \\ &\quad + \hat{R}_n(f^*) - R(f^*). \end{aligned}$$

Since $\hat{f}$ minimizes empirical risk,

$$\hat{R}_n(\hat{f}) - \hat{R}_n(f^*) \leq 0.$$

Hence

$$R(\hat{f}) - R(f^*) \leq 2 \sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)|.$$

Thus a uniform LLN implies consistency of ERM.

Rademacher Generalization Bound

Let

$$\mathcal{L} = \{(x, y) \mapsto \ell(f(x), y) : f \in \mathcal{F}\}.$$

Then

$$\sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)| = \sup_{g \in \mathcal{L}} |(P_n - P)g|.$$

Symmetrization gives

$$\mathbb{E} \sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)| \leq 2\mathfrak{R}_n(\mathcal{L}).$$

Therefore

$$\boxed{\mathbb{E}[R(\hat{f}) - R(f^*)] \leq 4\mathfrak{R}_n(\mathcal{L}).}$$

Empirical-process complexity directly controls prediction error.

Example: Linear Predictors

Consider

$$\mathcal{F} = \{x \mapsto w^\top x : \|w\|_2 \leq B\},$$

with

$$\|X_i\|_2 \leq R.$$

Then

$$\begin{aligned}\hat{\mathfrak{R}}_n(\mathcal{F}) &= \frac{1}{n} \mathbb{E}_\varepsilon \sup_{\|w\| \leq B} \left| w^\top \sum_i \varepsilon_i X_i \right| \\ &= \frac{B}{n} \mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i X_i \right\|_2.\end{aligned}$$

By Jensen,

$$\mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i X_i \right\|_2 \leq \left(\sum_i \|X_i\|_2^2 \right)^{1/2}.$$

Hence

$$\boxed{\hat{\mathfrak{R}}_n(\mathcal{F}) \leq \frac{BR}{\sqrt{n}}}.$$

Why This Is Interesting

The class

$$\mathcal{F} = \{x \mapsto w^\top x : \|w\| \leq B\}$$

contains infinitely many functions.

Yet

$$\mathfrak{R}_n(\mathcal{F}) \lesssim \frac{BR}{\sqrt{n}}.$$

So cardinality is not the right notion of complexity.

Remark:

An infinite-dimensional-looking optimization problem can still have small statistical complexity because the geometry of the parameter space constrains the possible predictions.

This motivates geometric and combinatorial complexity measures.

Now consider binary-valued functions

$$\mathcal{H} \subseteq \{0, 1\}^{\mathcal{X}}.$$

Given points

$$x_1, \dots, x_m,$$

consider the set of labelings

$$\mathcal{H}|_{x_{1:m}} = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\}.$$

The largest possible number of labelings is

$$2^m.$$

Can $\mathcal{H}$ realize all of them?

Shattering and VC Dimension

Definition:

A set $\{x_1, \dots, x_m\}$ is *shattered* by $\mathcal{H}$ if

$$\mathcal{H}|_{x_{1:m}} = \{0, 1\}^m.$$

Definition:

The VC dimension is

$$VC(\mathcal{H}) = \max\{m : \text{some } m \text{ points are shattered}\}.$$

Thus VC dimension asks:

What is the largest sample on which the class can implement every possible binary labeling?

Example: Thresholds

Consider

$$\mathcal{H} = \{x \mapsto \mathbb{1}\{x \leq t\} : t \in \mathbb{R}\}.$$

One point can be shattered.

For two points

$$x_1 < x_2,$$

possible labelings include

$$(0, 0), \quad (1, 0), \quad (1, 1),$$

but not

$$(0, 1).$$

Therefore

$$\boxed{\text{VC}(\mathcal{H}) = 1.}$$

This is exactly the class underlying the empirical CDF.

Example: Intervals

Consider

$$\mathcal{H} = \{x \mapsto \mathbb{1}\{x \in [a, b]\} : a \leq b\}.$$

Any two points can be shattered.

For three ordered points

$$x_1 < x_2 < x_3,$$

the labeling

$$(1, 0, 1)$$

cannot be realized by a single interval.

Therefore

$$\boxed{\text{VC}(\mathcal{H}) = 2.}$$

Example: Linear Classification

Consider affine halfspaces in $\mathbb{R}^d$:

$$\mathcal{H} = \{x \mapsto \mathbb{1}\{w^\top x + b \geq 0\}\}.$$

Then

$$\boxed{\text{VC}(\mathcal{H}) = d + 1.}$$

Examples:

Class	VC dimension
thresholds on $\mathbb{R}$	1
intervals on $\mathbb{R}$	2
affine halfspaces in $\mathbb{R}^2$	3
affine halfspaces in $\mathbb{R}^d$	$d + 1$

VC dimension turns model geometry into a combinatorial quantity.

The Growth Function

Definition:

The growth function is

$$\Pi_{\mathcal{H}}(n) = \sup_{x_1, \dots, x_n} |\mathcal{H}|_{x_{1:n}}|.$$

Always

$$\Pi_{\mathcal{H}}(n) \leq 2^n.$$

If $\mathcal{H}$ shatters n points,

$$\Pi_{\mathcal{H}}(n) = 2^n.$$

If

$$d = \text{VC}(\mathcal{H}) < \infty,$$

then exponential growth cannot continue after d .

How fast can $\Pi_{\mathcal{H}}(n)$ grow?

Theorem (Sauer–Shelah)

If

$$VC(\mathcal{H}) = d < \infty,$$

then for $n \geq d$,

$$\Pi_{\mathcal{H}}(n) \leq \sum_{j=0}^d \binom{n}{j} \leq \left(\frac{en}{d}\right)^d.$$

Therefore

$$\log \Pi_{\mathcal{H}}(n) \lesssim d \log \left(\frac{en}{d}\right).$$

Remark:

Finite VC dimension converts the number of possible labelings from potentially exponential in n to polynomial in n .

VC Dimension $\Rightarrow$ Uniform Convergence

Conditional on $X_1, \dots, X_n$, only

$$\Pi_{\mathcal{H}}(n)$$

different vectors

$$(h(X_1), \dots, h(X_n))$$

can occur.

Thus the finite-class Rademacher bound gives

$$\hat{\mathfrak{R}}_n(\mathcal{H}) \lesssim \sqrt{\frac{\log \Pi_{\mathcal{H}}(n)}{n}}.$$

By Sauer–Shelah,

$$\mathfrak{R}_n(\mathcal{H}) \lesssim \sqrt{\frac{d \log(en/d)}{n}}.$$

Hence finite VC dimension gives a uniform LLN.

VC Dimension and Generalization

For binary classification,

$$R(h) = P(h(X) \neq Y)$$

and

$$\hat{R}_n(h) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{h(X_i) \neq Y_i\}.$$

If $VC(\mathcal{H}) = d$, then roughly

$$\sup_{h \in \mathcal{H}} |\hat{R}_n(h) - R(h)| \lesssim \sqrt{\frac{d}{n}}$$

up to logarithmic factors in the elementary bound.

Therefore

$$R(\hat{h}) - \inf_{h \in \mathcal{H}} R(h)$$

is controlled by model complexity.

more expressive class $\iff$ larger potential generalization error.

Beyond VC Dimension: Metric Complexity

VC dimension is especially useful for indicator classes.

For general real-valued functions, we need a geometric notion of complexity.

Definition:

An ε -net for $(\mathcal{F}, d)$ is a finite set

$$\{f_1, \dots, f_N\}$$

such that every $f \in \mathcal{F}$ satisfies

$$d(f, f_j) \leq \varepsilon$$

for some j .

The smallest possible N is

$$N(\varepsilon, \mathcal{F}, d),$$

the *covering number*.

Metric Entropy

The quantity

$$\log N(\varepsilon, \mathcal{F}, d)$$

is called the *metric entropy*.

Interpretation:

$N(\varepsilon, \mathcal{F}, d)$ = number of functions distinguishable at resolution ε .

For a finite class,

$$N(\varepsilon, \mathcal{F}, d) \leq |\mathcal{F}|.$$

For an infinite class, the effective size depends on the scale ε .

Remark:

VC dimension measures combinatorial richness.

Covering numbers measure geometric richness.

The Empirical L_2 Metric

A natural metric for empirical processes is

$$d_n(f, g) = \|f - g\|_{P_n, 2} = \left[\frac{1}{n} \sum_{i=1}^n \{f(X_i) - g(X_i)\}^2 \right]^{1/2}.$$

Two functions are close if they give similar values on the observed sample.

Notice that d_n is random.

Remark:

This is useful rather than problematic: the complexity measure adapts to the observed design points.

We will control

$$N(\varepsilon, \mathcal{F}, L_2(P_n)).$$

Why One Covering Scale Is Not Enough

Take an ε -net $\mathcal{F}_\varepsilon$.

For each f , choose

$$\pi_\varepsilon f \in \mathcal{F}_\varepsilon.$$

Then

$$Z_f = Z_{\pi_\varepsilon f} + \{Z_f - Z_{\pi_\varepsilon f}\}.$$

There is a tradeoff:

	$ \mathcal{F}_\varepsilon $	approximation error
ε large	small	large
ε small	large	small

A single resolution cannot optimally balance both.

Solution: use all resolutions simultaneously.

The Chaining Idea

Choose

$$\varepsilon_0 > \varepsilon_1 > \varepsilon_2 > \cdots \downarrow 0$$

and an ε_j -net $\mathcal{F}_j$ at every scale.

For each f , let $\pi_j f$ be its approximation in $\mathcal{F}_j$.

Then

$$f = \pi_0 f + \sum_{j \geq 1} (\pi_j f - \pi_{j-1} f).$$

For a stochastic process,

$$Z_f - Z_{\pi_0 f} = \sum_{j \geq 1} (Z_{\pi_j f} - Z_{\pi_{j-1} f}).$$

Each increment is small, and at each scale there are only finitely many possibilities.

This is *chaining*.

Sub-Gaussian Increments

Definition:

A process $\{Z_f : f \in \mathcal{F}\}$ has sub-Gaussian increments under d if

$$\mathbb{E} e^{\lambda(Z_f - Z_g)} \leq \exp \left\{ \frac{\lambda^2 d(f, g)^2}{2} \right\}.$$

This implies

$$P(|Z_f - Z_g| > t) \lesssim \exp \left(-\frac{ct^2}{d(f, g)^2} \right).$$

Thus

$$d(f, g)$$

measures the typical size of the stochastic increment

$$Z_f - Z_g.$$

Geometry of the index set and probability of the process become linked.

Dudley's Entropy Bound

Theorem (Dudley)

For a centered process with sub-Gaussian increments,

$$\mathbb{E} \sup_{f \in \mathcal{F}} |Z_f - Z_{f_0}| \lesssim \int_0^{\text{diam}(\mathcal{F})} \sqrt{\log N(\varepsilon, \mathcal{F}, d)} d\varepsilon.$$

The integral

$$\int \sqrt{\log N(\varepsilon, \mathcal{F}, d)} d\varepsilon$$

accumulates complexity across all resolutions.

Remark:

A union bound works at one scale.
Chaining combines union bounds over a hierarchy of scales.

Where Does the Entropy Integral Come From?

Take dyadic scales

$$\varepsilon_j = 2^{-j} D.$$

At scale j ,

$$|\mathcal{F}_j| \leq N(\varepsilon_j, \mathcal{F}, d).$$

The increment

$$Z_{\pi_j f} - Z_{\pi_{j-1} f}$$

has size about ε_j .

The maximum over the net contributes roughly

$$\varepsilon_j \sqrt{\log N(\varepsilon_j, \mathcal{F}, d)}.$$

Summing:

$$\sum_j \varepsilon_j \sqrt{\log N(\varepsilon_j, \mathcal{F}, d)}.$$

This Riemann-sum-like expression becomes

$$\int_0^D \sqrt{\log N(\varepsilon, \mathcal{F}, d)} d\varepsilon.$$

Chaining the Rademacher Process

Conditional on the data, define

$$R_f = \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i).$$

Then

$$R_f - R_g = \frac{1}{n} \sum_i \varepsilon_i \{f(X_i) - g(X_i)\}.$$

Its MGF satisfies

$$\mathbb{E}_\varepsilon e^{\lambda(R_f - R_g)} \leq \exp \left\{ \frac{\lambda^2}{2n} \|f - g\|_{P_{n,2}}^2 \right\}.$$

Thus R_f is sub-Gaussian under

$$d(f, g) = \frac{1}{\sqrt{n}} \|f - g\|_{P_{n,2}}.$$

Dudley Bound for Rademacher Complexity

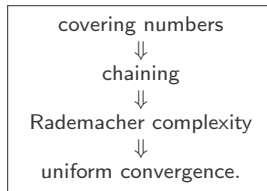
Applying chaining,

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \lesssim \frac{1}{\sqrt{n}} \int_0^{D_n} \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(P_n))} d\varepsilon,$$

where

$$D_n = \text{diam}(\mathcal{F}, L_2(P_n)).$$

So we have the chain



VC Classes Have Small Entropy

For a bounded VC class with VC dimension d ,

$$N(\varepsilon, \mathcal{F}, L_2(Q)) \lesssim \left(\frac{A}{\varepsilon}\right)^{Cd}.$$

Therefore

$$\log N(\varepsilon, \mathcal{F}, L_2(Q)) \lesssim d \log(A/\varepsilon).$$

Dudley's integral becomes

$$\int_0^1 \sqrt{d \log(A/\varepsilon)} d\varepsilon \lesssim \sqrt{d}.$$

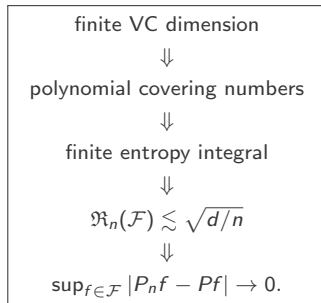
Hence

$$\boxed{\mathfrak{R}_n(\mathcal{F}) \lesssim \sqrt{\frac{d}{n}}}.$$

Chaining removes the extra $\sqrt{\log n}$ from the elementary Sauer–Shelah argument.

The Complexity Pipeline

For VC-type classes:



This gives a general route from combinatorial structure to statistical consistency.

Now Change the Scale

So far we studied

$$(P_n - P)f$$

and asked whether it vanishes uniformly.

Now consider

$$\mathbb{G}_n f = \sqrt{n}(P_n - P)f.$$

For a fixed f ,

$$\mathbb{G}_n f \Rightarrow N(0, \sigma_f^2),$$

where

$$\sigma_f^2 = P(f - Pf)^2.$$

For finitely many

$$f_1, \dots, f_k,$$

the multivariate CLT gives

$$(\mathbb{G}_n f_1, \dots, \mathbb{G}_n f_k) \Rightarrow N_k(0, \Sigma).$$

Can the *entire process* converge?

Definition:

A class $\mathcal{F}$ is *P-Donsker* if

$$\mathbb{G}_n \Rightarrow \mathbb{G}_P \quad \text{in } \ell^\infty(\mathcal{F}),$$

where $\mathbb{G}_P$ is a tight centered Gaussian process.

Its covariance is

$$\text{Cov}(\mathbb{G}_P f, \mathbb{G}_P g) = P(fg) - P f P g.$$

Compare:

$$\begin{array}{ll} \text{GC} & : \quad P_n - P \rightarrow 0 \text{ uniformly,} \\ \text{Donsker} & : \quad \sqrt{n}(P_n - P) \Rightarrow \text{Gaussian process.} \end{array}$$

Why Is Donsker Harder Than a CLT?

For any fixed finite collection

$$f_1, \dots, f_k,$$

the multivariate CLT already gives the correct limit.

So a functional CLT requires two ingredients:

1. finite-dimensional convergence,
2. tightness / asymptotic equicontinuity.

The second condition prevents the process from oscillating too wildly between nearby functions.

This is exactly where

covering numbers + chaining

enter again.

The Classical Donsker Theorem

Take again

$$\mathcal{F} = \{\mathbb{1}_{(-\infty, t]} : t \in \mathbb{R}\}.$$

Then

$$\mathbb{G}_n f_t = \sqrt{n}\{F_n(t) - F(t)\}.$$

Theorem (Donsker)

If F is continuous,

$$\sqrt{n}(F_n - F) \Rightarrow \mathbb{B} \circ F,$$

where $\mathbb{B}$ is a standard Brownian bridge.

Thus the empirical CDF has a limiting *random function*, not just a limiting random variable.

The Brownian Bridge

A standard Brownian bridge

$$\{\mathbb{B}(u) : 0 \leq u \leq 1\}$$

is a centered Gaussian process with covariance

$$\text{Cov}(\mathbb{B}(s), \mathbb{B}(t)) = \min(s, t) - st.$$

For the empirical process,

$$\begin{aligned} \text{Cov}(\mathbb{1}\{X \leq s\}, \mathbb{1}\{X \leq t\}) \\ = F(s \wedge t) - F(s)F(t). \end{aligned}$$

This is precisely

$$\text{Cov}(\mathbb{B}(F(s)), \mathbb{B}(F(t))).$$

Hence the Brownian bridge is the natural Gaussian limit.

Glivenko–Cantelli versus Donsker

For the empirical CDF:

$$\textbf{Glivenko–Cantelli:} \quad \sup_t |F_n(t) - F(t)| \rightarrow 0.$$

This tells us the estimator is uniformly consistent.

$$\textbf{Donsker:} \quad \sqrt{n}(F_n - F) \Rightarrow \mathbb{B} \circ F.$$

This describes the stochastic fluctuations around F .

GC gives consistency;	Donsker gives inference.
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Application: Kolmogorov–Smirnov Statistic

Define

$$D_n = \sup_t |F_n(t) - F(t)|.$$

Glivenko–Cantelli gives

$$D_n \rightarrow 0.$$

But for testing we need the scale of the fluctuations.

Donsker gives

$$\sqrt{n}D_n = \sup_t |\sqrt{n}(F_n(t) - F(t))| \Rightarrow \sup_{0 \leq u \leq 1} |\mathbb{B}(u)|.$$

Thus the functional CLT immediately yields the limiting distribution of a statistic involving a supremum over infinitely many t .

Application: Quantiles

Let

$$q_\alpha = F^{-1}(\alpha), \quad \hat{q}_\alpha = F_n^{-1}(\alpha).$$

GC gives

$$\hat{q}_\alpha \rightarrow q_\alpha$$

under regularity.

If F has density f with

$$f(q_\alpha) > 0,$$

then

$$\sqrt{n}(\hat{q}_\alpha - q_\alpha) = -\frac{\sqrt{n}\{F_n(q_\alpha) - \alpha\}}{f(q_\alpha)} + o_P(1).$$

Therefore

$$\boxed{\sqrt{n}(\hat{q}_\alpha - q_\alpha) \Rightarrow N\left(0, \frac{\alpha(1-\alpha)}{f(q_\alpha)^2}\right)}.$$

A General Empirical-Process Template

Many asymptotic-statistics arguments have the form

empirical process + linearization + continuous mapping



asymptotic distribution of an estimator.

Examples include:

- quantiles,
- M-estimators,
- Z-estimators,
- goodness-of-fit statistics,
- semiparametric estimators.

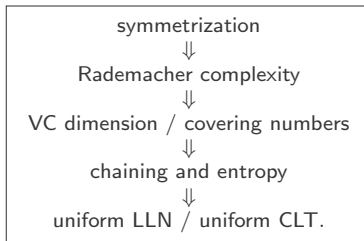
This is why empirical-process theory becomes central in modern asymptotic statistics.

Week 6: Main Picture

Starting from

$$P_n f - P f,$$

we developed the chain



The same machinery appears in both

asymptotic statistics and statistical learning theory.

Takeaways

- For a fixed f ,

$$P_n f - P f$$

is governed by ordinary LLN/CLT theory.

- For an entire class $\mathcal{F}$, we must control its complexity.
- Rademacher complexity measures the ability to fit random noise.
- VC dimension controls combinatorial complexity.
- Covering numbers measure complexity at resolution ε .
- Chaining combines information across all resolutions.
- Glivenko–Cantelli classes satisfy a uniform LLN.
- Donsker classes satisfy a uniform CLT.

Complexity determines whether pointwise probability results become uniform.